

Molecular structure inference from molecular spectra

with chembedding and functional groups

NMR annotation to multitask estimation
& learning equivalent SMILES sequences

Training setup:

NMR annotation $\rightarrow$  SMILES 

where

- , ,  are task-specific trigger tokens,
-  is where the hidden state is supervised,
- SMILES is the next-token distribution supervised serialization,
-  is the supervised end-of-sequence token.

Losses:

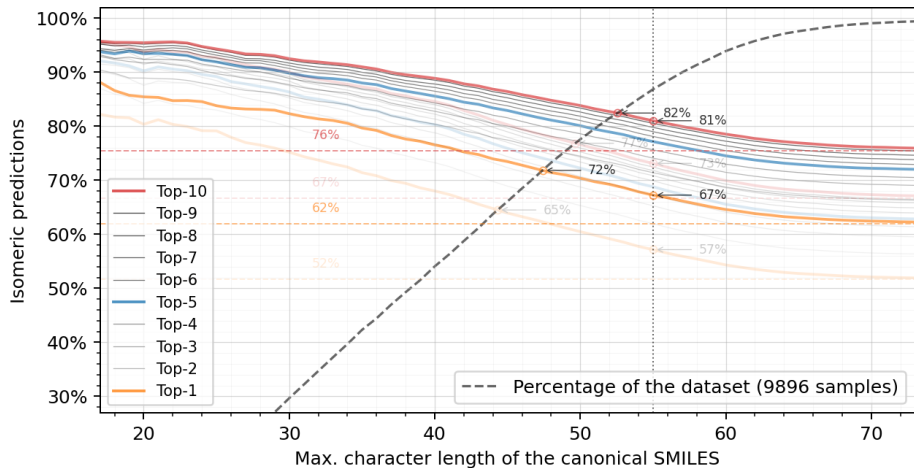
- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log \det T + \frac{n}{2} \log 2\pi)$ – chembedding ($n = 384$)
- $\frac{1}{m}(\|S\mathbf{e}\|_1 - \log \det S + m \log 2)$ – functional groups ($m = 289$)
- KL divergence, position-averaged – next-token distribution
- Causal LM = $-\log p(\text{eos} \mid \text{prefix})$ – end-of-sequence token

with trainable matrices T and S , combined à la Cipolla et al. [1]:

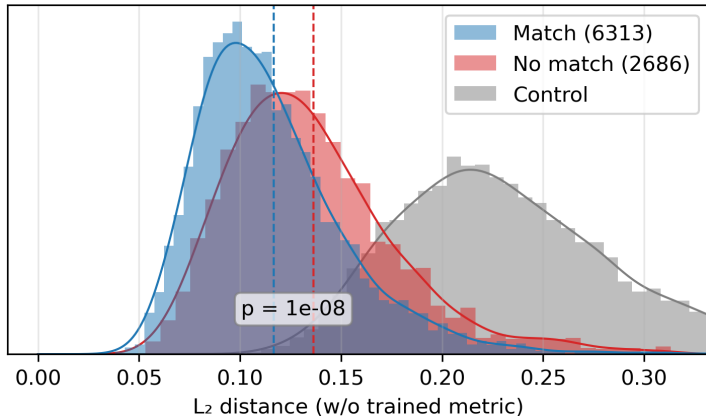
$$\text{minimize} \quad \sum_t \left(\frac{1}{\sigma_t^2} \text{softplus}(\text{loss}_t) + \log \sigma_t^2 \right)$$

with a trainable σ_t for each task t .

Accuracy of structure inference

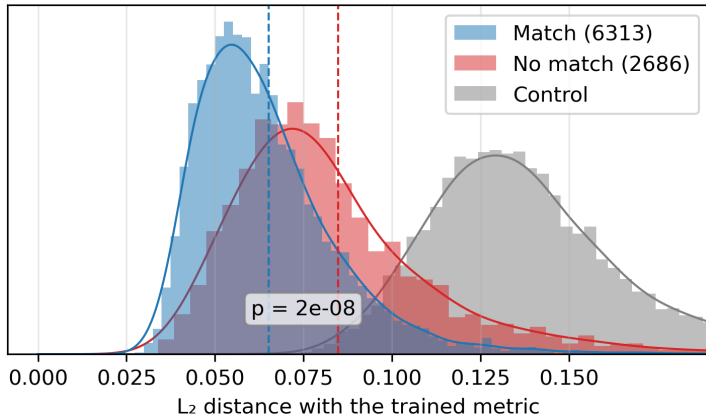


Accuracy of the “chemembedding”



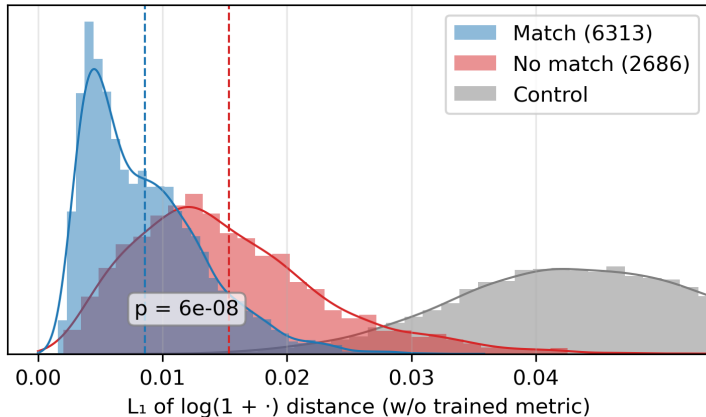
Distance between the predicted “chemembedding” and that of the canonical SMILES of the target molecule across a random subset of the test set.

Accuracy of the “chembedding”



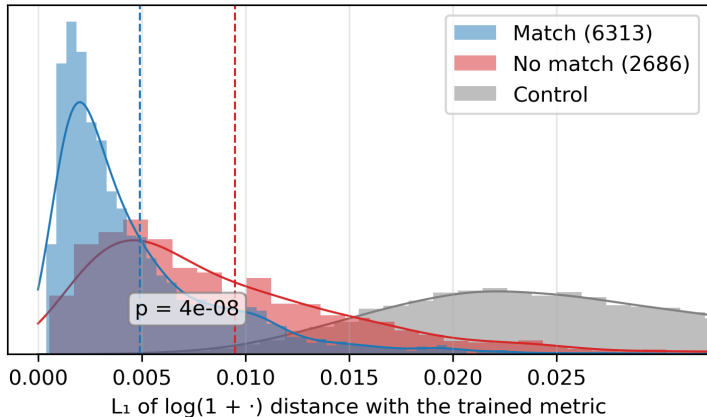
Distance between the predicted “chembedding” and that of the canonical SMILES of the target molecule across a random subset of the test set.

Accuracy of functional group counts



Distance between the predicted functional group $\log(1 + \text{counts})$ and those of the target molecule across a random subset of the test set.

Accuracy of functional group counts



Distance between the predicted functional group $\log(1 + \text{counts})$ and those of the target molecule across a random subset of the test set.

thx.

References I

- [1] Roberto Cipolla, Yarin Gal, and Alex Kendall. “Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics”. In: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2018). [url].

